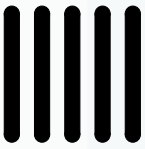


Optimization
MAT 3007

Linear Algebra — Cheat Sheet



Andre Milzarek, Junfeng Wu, Junchi Yang
The Chinese University of Hong Kong, Shenzhen
SDS · School of Data Science

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Remarks and Versions

The purpose of this handout is to give a brief review of some of the basic concepts in linear algebra. If you are unfamiliar with the material and/or would like to do some further reading, you may consult, e.g., the books:

- Strang, G.: *Introduction to Linear Algebra* (Vol. 3). Wellesley, MA: Wellesley-Cambridge Press, (1993).
- Horn, R. A., and Johnson, C. R.: *Matrix analysis*. Cambridge University Press, (2012).
- Axler, S. J.: *Linear algebra done right* (Vol. 2). New York: Springer, (1997).

Version. This is version v1.0 (date: 2025/01/15).

1 Vector Spaces

We denote the set of real numbers (also referred to as scalars) by $\mathbb{R}$. A vector space that we may be familiar with is $\mathbb{R}^2$. We can think of it as a plane, and any point in $\mathbb{R}^2$ can be represented by an ordered list of its coordinates, that is,

$$\mathbb{R}^2 = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x, y \in \mathbb{R} \right\}.$$

To define n-dimensional analogues of $\mathbb{R}^2$, where n is a positive integer, we will replace $\mathbb{R}$ with $\mathbb{F}$ ($\mathbb{F}$ is a *field*, which could be the set $\mathbb{R}$ of real numbers or the set $\mathbb{C}$ of complex numbers) and replace 2 with n. Therefore, an n-dimensional vector will be represented as

$$\mathbb{F}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_i \in \mathbb{F} \text{ for } i = 1, \ldots, n \right\}.$$

We can easily define algebraic manipulations on $\mathbb{F}^n$. For example, the addition of two elements in $\mathbb{F}^n$ is defined by adding corresponding coordinates:

$$\boldsymbol{x} + \boldsymbol{y} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

and the scalar multiplication on an element of $\mathbb{F}^n$ is defined by performing multiplication in each coordinate:

$$a\boldsymbol{x} = a \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} ax_1 \\ \vdots \\ ax_n \end{bmatrix},$$

where $a \in \mathbb{F}$. The motivation for the definition of a vector space is from the properties possessed by addition and scalar multiplication operations on $\mathbb{F}^n$. Formally, we say that a set $\mathcal{V}$ is a vector space over a field $\mathbb{F}$, if there is a binary operation "+", called addition, in $\mathcal{V}$, and a map $\mathbb{F} \times \mathcal{V} \to \mathcal{V}$, called scalar multiplication (we typically write "$\cdot$"), such that the following properties hold:

- *Commutativity*: $\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{y} + \boldsymbol{x}$ for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$;
- *Associativity*: $(\boldsymbol{x} + \boldsymbol{y}) + \boldsymbol{z} = \boldsymbol{x} + (\boldsymbol{y} + \boldsymbol{z})$ and $(ab) \cdot \boldsymbol{x} = a(b \cdot \boldsymbol{x})$ for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ and $a, b \in \mathbb{F}$;
- *Additive Identity*: There exists an element $\mathbf{0} \in \mathcal{V}$ such that $\boldsymbol{x} + \mathbf{0} = \boldsymbol{x}$ for all $\boldsymbol{x} \in \mathcal{V}$;
- *Additive Inverse*: For any $\boldsymbol{x} \in \mathcal{V}$, there exists an element $\boldsymbol{y} \in \mathcal{V}$ such that $\boldsymbol{x} + \boldsymbol{y} = \mathbf{0}$;
- *Multiplicative Identity*: $1 \cdot \boldsymbol{x} = \boldsymbol{x}$ for all $\boldsymbol{x} \in \mathcal{V}$;
- *Distributive Properties*: $a \cdot (\boldsymbol{x} + \boldsymbol{y}) = a \cdot \boldsymbol{x} + a \cdot \boldsymbol{y}$ and $(a + b) \cdot \boldsymbol{x} = a \cdot \boldsymbol{x} + b \cdot \boldsymbol{x}$ for all $a, b \in \mathbb{F}$ and $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$.

An element of a vector space is called a *vector*. A subset $\mathcal{U}$ of $\mathcal{V}$ is called a subspace of $\mathcal{V}$ if $\mathcal{U}$ itself is also a vector space.

In what follows, we are mainly interested in vector spaces over the field $\mathbb{R}$.

2 Linear Independence and Bases

We say that a finite collection $C = \{\boldsymbol{x}^1, \boldsymbol{x}^2, \ldots, \boldsymbol{x}^m\}$ of vectors in $\mathbb{R}^n$ is *linearly dependent* if there exist scalars $a_1, \ldots, a_m \in \mathbb{R}$, not all of them are zero, such that $\sum_{i=1}^m a_i \boldsymbol{x}^i = \boldsymbol{0}$. The collection $C = \{\boldsymbol{x}^1, \boldsymbol{x}^2, \ldots, \boldsymbol{x}^m\}$ is said to be *linearly independent* if it is *not* linearly dependent.

A linear combination of a collection $\{\boldsymbol{x}^1, \boldsymbol{x}^2, \ldots, \boldsymbol{x}^m\}$ of vectors in $\mathbb{R}^n$ is a vector of the form

$$a_1 \boldsymbol{x}^1 + \cdots + a_m \boldsymbol{x}^m,$$

where $a_1, \ldots, a_m \in \mathbb{R}$. The set of all linear combinations of $\{\boldsymbol{x}^1, \boldsymbol{x}^2, \ldots, \boldsymbol{x}^m\}$ is called the *span* of $\{\boldsymbol{x}^1, \boldsymbol{x}^2, \ldots, \boldsymbol{x}^m\}$. The span of a collection of vectors in $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n$.

A *basis* of $\mathbb{R}^n$ is a collection of vectors in $\mathbb{R}^n$ that is linearly independent and spans the whole space $\mathbb{R}^n$. For example,

$$\{\boldsymbol{x}^1, \boldsymbol{x}^2\}, \quad \text{where} \quad \boldsymbol{x}^1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \boldsymbol{x}^2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

is a basis of $\mathbb{R}^2$, and

$$\{\boldsymbol{e}^1, \boldsymbol{e}^2\}, \quad \text{where} \quad \boldsymbol{e}^1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{e}^2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

is a standard basis of $\mathbb{R}^2$.

3 Vectors and Norms

As introduced in the first section, a *point* or *vector* $\boldsymbol{x} \in \mathbb{R}^n$ is represented by a column vector

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \text{with components} \quad x_i \in \mathbb{R}, \quad \forall\, i.$$

The *transposed* $\boldsymbol{x}^\top$ of a vector $\boldsymbol{x} \in \mathbb{R}^n$ is the row vector $\boldsymbol{x}^\top = [x_1, x_2, \ldots, x_n]$.

For $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$, we define the *Euclidean inner product* by

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{x}^\top \boldsymbol{y} = \sum\nolimits_{i=1}^n x_i y_i.$$

The inner product is a function $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ with the following properties:

- Positivity: $\langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq 0$ and $\langle \boldsymbol{x}, \boldsymbol{x} \rangle = 0$ if and only if $\boldsymbol{x} = \boldsymbol{0}$,
- Symmetry: $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle$,

- Linearity: $\langle \alpha \boldsymbol{x} + \beta \boldsymbol{y}, \boldsymbol{z} \rangle = \alpha \langle \boldsymbol{x}, \boldsymbol{z} \rangle + \beta \langle \boldsymbol{y}, \boldsymbol{z} \rangle$,

for all $\alpha, \beta \in \mathbb{R}$ and $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^n$. It is possible to define other functions on $\mathbb{R}^n \times \mathbb{R}^n$ that satisfy the latter properties and many results involving the Euclidean inner product also hold for these other forms of inner products.

Inner products allow to measure angles and distances between vectors. Two vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ are called *orthogonal* if $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{x}^\top \boldsymbol{y} = 0$.

The *Euclidean norm* of a point $\boldsymbol{x} \in \mathbb{R}^n$ is defined as:

$$\|\boldsymbol{x}\|_2 = \|\boldsymbol{x}\| = \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle} = \sqrt{\boldsymbol{x}^\top \boldsymbol{x}} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

The Euclidean norm satisfies the following properties for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$:

- Positivity: $\|\boldsymbol{x}\| \geq 0$ and $\|\boldsymbol{x}\| = 0$ if and only if $\boldsymbol{x} = \boldsymbol{0}$.
- Homogeneity: $\|\alpha \boldsymbol{x}\| = |\alpha| \|\boldsymbol{x}\|$.
- Triangle inequality: $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$.
- *Cauchy-Schwarz inequality*: $|\langle \boldsymbol{x}, \boldsymbol{y} \rangle| \leq \|\boldsymbol{x}\| \|\boldsymbol{y}\|$. Equality holds if and only if $\boldsymbol{x}$ and $\boldsymbol{y}$ are parallel (i.e., there exists $\alpha \in \mathbb{R}$ such that $\boldsymbol{x} = \alpha \boldsymbol{y}$).

We briefly verify the last (important) property:

Proof. If $\boldsymbol{x} = \boldsymbol{0}$ or $\boldsymbol{y} = \boldsymbol{0}$, the inequality is obviously satisfied. Thus, we can assume $\boldsymbol{x}, \boldsymbol{y} \neq \boldsymbol{0}$ without loss of generality. Let us set $\tilde{\boldsymbol{x}} = \boldsymbol{x}/\|\boldsymbol{x}\|$ and $\tilde{\boldsymbol{y}} = \boldsymbol{y}/\|\boldsymbol{y}\|$, then we have

$$0 \leq \|\tilde{\boldsymbol{x}} - \tilde{\boldsymbol{y}}\|^2 = \langle \tilde{\boldsymbol{x}} - \tilde{\boldsymbol{y}}, \tilde{\boldsymbol{x}} - \tilde{\boldsymbol{y}} \rangle = \|\tilde{\boldsymbol{x}}\|^2 - 2\langle \tilde{\boldsymbol{x}}, \tilde{\boldsymbol{y}} \rangle + \|\tilde{\boldsymbol{y}}\|^2 = 2 - 2\langle \tilde{\boldsymbol{x}}, \tilde{\boldsymbol{y}} \rangle,$$

which implies $\langle \tilde{\boldsymbol{x}}, \tilde{\boldsymbol{y}} \rangle \leq 1$ and $\langle \boldsymbol{x}, \boldsymbol{y} \rangle \leq \|\boldsymbol{x}\| \|\boldsymbol{y}\|$. Moreover, equality only holds in the case $\tilde{\boldsymbol{x}} = \tilde{\boldsymbol{y}}$. We now repeat the same calculation with $\tilde{\boldsymbol{x}} = -\boldsymbol{x}/\|\boldsymbol{x}\|$ which yields $-\langle \boldsymbol{x}, \boldsymbol{y} \rangle \leq \|\boldsymbol{x}\| \|\boldsymbol{y}\|$. This establishes the Cauchy-Schwarz inequality. Equality is only satisfied if $\boldsymbol{x}/\|\boldsymbol{x}\| = \pm \boldsymbol{y}/\|\boldsymbol{y}\|$, i.e., if $\boldsymbol{x} = \alpha \boldsymbol{y}$ for some $\alpha \in \mathbb{R}$. ■

A mapping $r : \mathbb{R}^n \to \mathbb{R}$ satisfying the first three mentioned properties is said to be a *norm* on $\mathbb{R}^n$. There are norms that can not be directly derived from an underlying inner product. The following functions are popular examples of norms:

- ℓ_1-norm: $\|\boldsymbol{x}\|_1 = \sum_{i=1}^n |x_i|$.
- Euclidean norm: $\|\boldsymbol{x}\|_2 = \left[\sum_{i=1}^n |x_i|^2\right]^{1/2}$.
- ℓ_p-norm or (Hölder) p-norm: $\|\boldsymbol{x}\|_p = [\sum_{i=1}^n |x_i|^p]^{1/p}$ where $p \in [1, \infty)$.
- Maximum norm: $\|\boldsymbol{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|$.

The notation $\|\cdot\|$ is often used to express a general norm function. Here – if not otherwise stated – we will always use the convention $\|\cdot\| = \|\cdot\|_2$.

In $\mathbb{R}^n$, it can be shown that all norms are *equivalent*. In particular, let $\|\cdot\|$ and $|||\cdot|||$ be

two different norms on $\mathbb{R}^n$. Then there exist constants $c, C > 0$ such that

$$c \cdot |||\boldsymbol{x}||| \leq \|\boldsymbol{x}\| \leq C \cdot |||\boldsymbol{x}|||, \quad \forall\, \boldsymbol{x} \in \mathbb{R}^n.$$

The constants c, C typically depend on the dimension n.

4 Linear Maps and Matrices

A linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a function $T : \mathbb{R}^n \to \mathbb{R}^m$, satisfying the following property:

$$T(\alpha\boldsymbol{x} + \beta\boldsymbol{y}) = \alpha T(\boldsymbol{x}) + \beta T(\boldsymbol{y}) \quad \text{for all} \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n \quad \text{and} \quad \alpha, \beta \in \mathbb{R}.$$

We denote the set of all linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$ as $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$. For $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ the *null space* $\text{null}(T)$ (also referred to as kernel) of T is the subset of $\mathbb{R}^n$ that consists of vectors mapped to $\mathbf{0}$ by T, i.e.,

$$\text{null}(T) := \{\boldsymbol{x} \in \mathbb{R}^n : T(\boldsymbol{x}) = \mathbf{0}\}.$$

The *range* of T, $\text{range}(T)$, is the subset of $\mathbb{R}^m$ consisting of vectors that are of the form $T(\boldsymbol{x})$ for some $\boldsymbol{x} \in \mathbb{R}^n$:

$$\text{range}(T) = \{\boldsymbol{y} \in \mathbb{R}^m : \exists\, \boldsymbol{x} \in \mathbb{R}^n \,\text{with}\, T(\boldsymbol{x}) = \boldsymbol{y}\}.$$

The null space of T is a subspace of $\mathbb{R}^n$ while the range is a subspace of $\mathbb{R}^m$.

We use $\mathbb{R}^{m\times n}$ to denote the set of $m \times n$ matrices (arrays) whose components are elements from $\mathbb{R}$. We can represent an element $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ as follows:

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad \text{with components} \quad a_{ij} \in \mathbb{R}, \quad \forall\, i, j. \tag{4.1}$$

For a linear map $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, suppose that $\{\boldsymbol{y}^1, \ldots, \boldsymbol{y}^n\}$ is a basis of $\mathbb{R}^n$ and let $\{\boldsymbol{x}^1, \ldots, \boldsymbol{x}^m\}$ be a basis of $\mathbb{R}^m$. For each $k = 1, \ldots, n$, suppose that we can write $T(\boldsymbol{y}^k)$ (uniquely) as a linear combination of $\boldsymbol{x}^1, \ldots, \boldsymbol{x}^m$ as follows:

$$T(\boldsymbol{y}^k) = a_{1k}\boldsymbol{x}^1 + \cdots + a_{mk}\boldsymbol{x}^m.$$

Then the scalars a_{jk}, $j = 1, \ldots, m$, $k = 1, \ldots, n$ completely determine the linear map T. We call the $m \times n$ array in (4.1) the *matrix* of T with respect to $\{\boldsymbol{y}^1, \ldots, \boldsymbol{y}^n\}$ and $\{\boldsymbol{x}^1, \ldots, \boldsymbol{x}^m\}$. A *row vector* is a matrix with $m = 1$ and a *column vector* is a matrix with $n = 1$.

Given an $m \times n$ matrix $\boldsymbol{A}$, the transposed matrix $\boldsymbol{A}^\top \in \mathbb{R}^{n\times m}$ is defined by swapping columns with rows of A. In particular, the i-th column of $\boldsymbol{A}^\top$ is set to be the i-th row of $\boldsymbol{A}$:

$$\boldsymbol{A}^\top = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix} \in \mathbb{R}^{n\times m}.$$

We often adopt the following row- and column-based perspectives for a given matrix $\boldsymbol{A}$:

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} \boldsymbol{A}_1 & \cdots & \boldsymbol{A}_n \end{bmatrix} = \begin{bmatrix} \boldsymbol{a}_1^\top \\ \vdots \\ \boldsymbol{a}_m^\top \end{bmatrix},$$

where $\boldsymbol{A}_j \in \mathbb{R}^m$ is the j-th column of $\boldsymbol{A}$ and $\boldsymbol{a}_i^\top$, $\boldsymbol{a}_i \in \mathbb{R}^n$, is the i-th row of $\boldsymbol{A}$. Given these representations and $\boldsymbol{x} \in \mathbb{R}^n$, we can compute

$$\boldsymbol{A}\boldsymbol{x} = \begin{bmatrix} \boldsymbol{A}_1 & \cdots & \boldsymbol{A}_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1\boldsymbol{A}_1 + x_2\boldsymbol{A}_2 + \cdots + x_n\boldsymbol{A}_n \quad \text{and} \quad \boldsymbol{A}\boldsymbol{x} = \begin{bmatrix} \boldsymbol{a}_1^\top \\ \vdots \\ \boldsymbol{a}_m^\top \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} \boldsymbol{a}_1^\top \boldsymbol{x} \\ \vdots \\ \boldsymbol{a}_m^\top \boldsymbol{x} \end{bmatrix}.$$

Hence, we have $\boldsymbol{y} \in \text{range}(\boldsymbol{A})$ if and only if $\boldsymbol{y}$ can be written as a linear combination of the n columns $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n$ of $\boldsymbol{A}$.

We continue with several basic notations and terminologies:

- A matrix is called *square* if $m = n$.
- A square matrix $\boldsymbol{A}$ is called *symmetric* if $\boldsymbol{A}^\top = \boldsymbol{A}$.
- The *identity matrix* is the square matrix $\boldsymbol{I}$ with $I_{ii} = 1$ for all i and $I_{ij} = 0$ for $i \neq j$.

A matrix $\boldsymbol{A}$ is called *orthogonal* if all of its columns are pair-wise orthogonal to each other. $\boldsymbol{A}$ is called *orthonormal* if $\boldsymbol{A}^\top \boldsymbol{A} = \boldsymbol{I}$. The *trace* of a square matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ is defined by

$$\text{tr}(\boldsymbol{A}) = a_{11} + a_{22} + \cdots + a_{nn}.$$

The matrix-matrix product between $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{B} \in \mathbb{R}^{n \times p}$ is defined as

$$\boldsymbol{C} = \boldsymbol{A}\boldsymbol{B} \in \mathbb{R}^{m \times p} \quad \text{where} \quad c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

It holds that $\boldsymbol{C}^\top = (\boldsymbol{A}\boldsymbol{B})^\top = \boldsymbol{B}^\top \boldsymbol{A}^\top$.

5 Rank and Matrix Inversion

The rank of a matrix $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, denoted by $\text{rank}(\boldsymbol{A})$, is defined as the largest possible number of linearly independent column vectors or row vectors. Some facts about the rank of a matrix:

- $\text{rank}(\boldsymbol{A}) = \text{rank}(\boldsymbol{A}^\top)$;
- $\text{rank}(\boldsymbol{A} + \boldsymbol{B}) \leq \text{rank}(\boldsymbol{A}) + \text{rank}(\boldsymbol{B})$;
- $\text{rank}(\boldsymbol{A}\boldsymbol{B}) \leq \min\{\text{rank}(\boldsymbol{A}), \text{rank}(\boldsymbol{B})\}$.

An $n \times n$ square matrix $\boldsymbol{A}$ is said to be *invertible* if $\text{rank}(\boldsymbol{A}) = n$. The inverse of the matrix $\boldsymbol{A}$ is denoted as $\boldsymbol{A}^{-1}$, and we have

$$\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}.$$

Here, we list some additional facts:

- $(\boldsymbol{A}^{-1})^{-1} = \boldsymbol{A}$.
- $(\boldsymbol{A}\boldsymbol{B})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$ where $\boldsymbol{A}$ and $\boldsymbol{B}$ are square, invertible matrices.
- $(\boldsymbol{A}^\top)^{-1} = (\boldsymbol{A}^{-1})^\top$.

6 Note on Hyperplanes

The notion of hyperplanes that is used in the support vector machine example in this course refers to an affine hyperplane of an n-dimensional Euclidean space $\mathbb{R}^n$. It is a higher-dimensional generalization of a line in $\mathbb{R}$ and a plane in $\mathbb{R}^2$.

Roughly speaking, an affine hyperplane in $\mathbb{R}^n$ is a flat subset with dimension $n-1$. In the Cartesian coordinates, any hyperplane, as a set of points, can be described with a single linear equation of the following form:

$$H := \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{a}^\top \boldsymbol{x} + b = 0\}, \tag{6.1}$$

where $\boldsymbol{a} \in \mathbb{R}^n \backslash \{\boldsymbol{0}\}$.

To derive a single linear equation in this expression, we denote by $\boldsymbol{v}$ a (nonzero) vector (called normal or support vector) that is normal to the hyperplane in question (i.e., $\boldsymbol{v}$ is perpendicular to any line in the hyperplane). Picking an arbitrary point $\boldsymbol{x}'$ in the hyperplane, we have the following equality for each point $\boldsymbol{x}$ in the hyperplane due to orthogonality:

$$\boldsymbol{v}^\top(\boldsymbol{x} - \boldsymbol{x}') = 0.$$

By setting $\boldsymbol{a} := \boldsymbol{v}$ and $b := -\boldsymbol{v}^\top \boldsymbol{x}'$, we obtain (6.1). Notice that, no matter what point $\boldsymbol{x}'$ is selected from the hyperplane, b is a constant (Why?).

Consider two parallel hyperplanes $H_1 := \{\boldsymbol{x} : \boldsymbol{a}^\top \boldsymbol{x} = b_1\}$ and $H_2 := \{\boldsymbol{x} : \boldsymbol{a}^\top \boldsymbol{x} = b_2\}$. The distance between the two hyperplanes can be computed by projecting any point $\boldsymbol{x}_1$ in the former hyperplane onto the latter one. Let the projection of $\boldsymbol{x}_1$ on H_2 be $\boldsymbol{x}_2$. Then the distance between H_1 and H_2 can be computed as $\|\boldsymbol{x}_2 - \boldsymbol{x}_1\|$. The vector $\boldsymbol{x}_2 - \boldsymbol{x}_1$ is orthogonal to H_2, that is, $\boldsymbol{x}_2 - \boldsymbol{x}_1 = \lambda \boldsymbol{a}$ for some $\lambda \in \mathbb{R}$. Substituting $\boldsymbol{x}_2 = \boldsymbol{x}_1 + \lambda \boldsymbol{a}$ into $\boldsymbol{a}^\top \boldsymbol{x}_2 = b_2$, we obtain

$$\boldsymbol{a}^\top \boldsymbol{x}_1 + \lambda \boldsymbol{a}^\top \boldsymbol{a} = \boldsymbol{b}_2.$$

Since $\boldsymbol{a}^\top \boldsymbol{x}_1 = b_1$, we have $b_2 - b_1 = \lambda \boldsymbol{a}^\top \boldsymbol{a} = \lambda \|\boldsymbol{a}\|^2$. Then $\|\boldsymbol{x}_2 - \boldsymbol{x}_1\| = \|\lambda \boldsymbol{a}\| = |b_2 - b_1|$, that is, the distance between H_1 and H_2 is $|b_2 - b_1| / \|\boldsymbol{a}\|$.